Tools — Shell

Tools for running shell commands and managing shell tasks. `Bash` commands pass through a 7-layer security model: safe-command whitelist, banned-command checks, denied patterns, directory-escape prevention, syntax validation, obfuscation detection, and user allowlist/denylist.

Tool	Description
`Bash`	Execute a shell command (7-layer security).
`bash_reset`	Reset persistent shell state.
`bg`	Manage shell tasks (spawn, list, wait, cancel).
`open`	Open a file/folder/URL via the desktop handler.

Use cases: running builds, tests, git operations, long-running tasks.

System instruction: "For simple commands use `Bash` with the `command` parameter immediately. Do not describe what you will run."

Bash

Execute a shell command and return its stdout and stderr. Commands run in the agent's working directory in a persistent shell.

Arguments

Argument	Type	Required	Description	Typical value
`command`	string	yes	Shell command to execute	`"cargo test"`
`timeout`	integer	no	Timeout in seconds	`120` (default); up to `1500` for builds/tests

Example:

```
Bash command="cargo test -- --nocapture" timeout=1500
Bash command="cargo fmt"
```

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bash_reset

Reset the persistent shell state for the current session, clearing saved working-directory changes and environment variables. No-op if no persistent state exists.

Arguments: none.

```
bash_reset
```

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bg

Manage **shell tasks**: spawn long-running commands that continue while the agent does other work, then inspect, follow, or stop them.

A shell task is not a sub-agent. Its rows live in the background-task store, in a different id space from the sub-agent task registry, so a shell task never appears in `list_agents` and can never be reached with `cancel_agent` or `wait_agents` - use `new_agent` to start a sub-agent and `cancel_agent` / `wait_agents` to control one.

Arguments

Argument	Type	Required	Description	Typical value
`action`	enum	yes	`spawn`, `list`, `status`, `output`, `tail`, `cancel`, `wait`, `cleanup`	`"spawn"`
`command`	string	yes for `spawn`	Shell command to run in background	`"cargo build"`
`task_id`	string	yes for `status`, `output`, `tail`, `cancel`, `wait`	Shell task id	`"bg-3"`
`lines`	integer	no	Lines returned by `tail`	`20`
`status`	enum	no	`list` filter: `running` / `completed` / `failed` / `cancelled`	—
`limit`	integer	no	Max tasks for `list`	`50`
`timeout`	integer	no	`wait` timeout in seconds	`60`
`working_dir`	string	no	Working directory for `spawn`	session cwd
`completed_only`	boolean	no	`cleanup`: only completed/failed/cancelled tasks	`true`
`older_than_minutes`	integer	no	`cleanup`: minimum task age	`60`
`session_id`	string	no	`list` / `cleanup` scope. It must name the calling session and then behaves exactly as omitting it; a foreign id is refused	`"sess-42"`

Session scope (FR-019, FR-037): `bg` reads and deletes only the calling session's shell tasks. `session_id` naming any other session is refused with `reason_code: wrong_session`, naming both sessions; the refusal deletes nothing. Omit the parameter, or pass the caller's own id - the two produce an identical result.

Reason codes: `unknown_action`, `parameter_not_accepted`, `service_unavailable`, `wrong_session`.

Example:

```
bg action="spawn" command="cargo build"
bg action="tail" task_id="bg-1" lines=20
bg action="wait" task_id="bg-1" timeout=300
```

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open

Open or reveal a file, folder, or URL using the desktop default handler (`xdg-open` on Linux, `open` on macOS, `start` on Windows). May fail in headless or sandboxed contexts.

Arguments

Argument	Type	Required	Description	Typical value
`target`	string	yes	File path, folder path, or URL	`"report.pdf"`
`action`	enum	no	`open` (default), `reveal` (open parent directory), `url` (validate and open a URL)	`"open"`

Example:

```
open target="docs/howtos/tools/pdf/file-operations.md.pdf"
```